SETH BRYANT

PhD, MSc, P.Eng.
Postdoctoral Research Associate
University of Wisconsin–Madison
Flood Risk Modelling, Earth Observation, and Research Software Engineering
bryant.seth@gmail.com | sbryant8@wisc.edu
github.com/cefect | ceflood.com | ORCID 0000-0002-8799-3919

EDUCATION

Doctor of Engineering 2024 University of Potsdam, Institute for Environmental Sciences and Geography; GFZ German Research Centre for Geosciences, Hydrology Section: DFG-RTG NatRiskChange.
Doctoral thesis: *Aggregation and Disaggregation in Flood Risk Models*. Full text.

M.Sc. 2019 University of Alberta, Edmonton, Alberta, Canada: Water Resources Engineering.
MSc thesis: *Accumulating Flood Risk*. Full text.

Visiting M.Sc. Student 2017 TU Delft, Delft, Netherlands: Civil Engineering

B.Sc. 2009 Montana State University, Bozeman, Montana, USA: Civil Engineering

ACADEMIC EXPERIENCE

2025 – current:	Postdoctoral Research Associate, University of Wisconsin–Madison, U.S.A. Research software engineer and scientist in Beth Tellman's lab, addressing flood impacts of adaptation and urban expansion with satellite observations.
2025 – 2026:	Postdoctoral Research Associate, University of Arizona, U.S.A. Research software engineer in Beth Tellman's lab, developing deep-learning based super-resolution of flood inundation grids.
2024 – 2025:	Visiting Researcher, Fathom, U.K. Research and software engineering under Oliver Wing, developing and implementing a flood hazard grid downscaler for a global pipeline.
2021 – 2024:	Doctoral/Affiliate Researcher, University of Potsdam and GFZ German Research Centre for Geosciences, Germany. Research and software development supervised by Heidi Kreibich and Bruno Merz.
2019 – 2021:	Research Affiliate, University of Alberta, Canada. Collaborative flood risk, remote sensing, and flood depth estimation research.
2016 – 2019:	Research Assistant, University of Alberta, Canada. MSc thesis research supervised by Evan Davies.
2017 – 2018:	Research Assistant, Deltares, Netherlands. River morphology and sedimentation research supervised by Erik Mosselman.

Expertise:
My expertise combines hydraulic engineering, flood-risk modeling, Earth observation, and research software development to connect regional predictions with local flood impacts. My research examines how terrain, infrastructure, spatial resolution, and nonlinear damage relationships affect estimates of inundation, exposure, and access to essential services. A central focus is identifying when added map detail improves decisions and when uncertainty in the inputs limits its value.

I develop terrain-conditioned downscaling methods and the floodSR toolkit, and am building USFloods through multisensor satellite mapping, deep learning, and scalable geospatial processing. These efforts connect physical constraints with observed flooding to reconstruct events, evaluate models, and quantify uncertainty. My developing research program extends this foundation to transferable flood forecasting, satellite-constrained floodplain storage, and downstream flood protection.

My consulting practice draws on 20 years of river and hydraulic engineering, flood hazard and risk assessment, and public-sector implementation. This includes FEMA flood mapping support, hydraulic design, elevation certifications, and development of CanFlood and RICorDE. Through CEFlood, I modernize CanFlood, develop CanCurve, and advise on depth-damage functions, average annual loss, model sensitivity, and uncertainty.

TECHNICAL SKILLS

Research and modelling:

- Flood risk analysis and modelling; GIS analysis and processing; hydraulic modelling; ML inference engineering; high-performance computing; grant application writing; data wrangling and analysis.

Programming:

- Python, pyQGIS, PostGIS, Git, LaTeX, Google Earth Engine, PyTorch, R, SSH, Unix shell, Windows batch scripting, Osmium Tool.

Software:

- QGIS, AutoCAD Civil 3D, Microsoft Office, HEC-RAS, SOBEK, HEC-HMS, Delft3D, GitHub/GitLab.

CURRENT RESEARCH PROJECTS

- Physics-guided super-resolution downscaling for national urban flood inundation mapping. (Natural Resources Canada Flood Hazard Identification and Mapping Program), $225,000, 2024.
- Co-I. "AI-Driven Assessment of U.S. Inland Flood Inundation at High Resolution across the 21st Century." 2026–2028. NCAR EdEc Faculty Innovation Program Cohort 4. Award Amount: $250,000. PI: Beth Tellman.
- Co-I. EAGER: Satellite-Based Evaluation of U.S. Flood-Mitigation Infrastructure Performance. NSF. Award Amount: $300,000. PI: Beth Tellman.

AWARDS AND SCHOLARSHIPS

- NSERC Postgraduate Scholarship–Doctoral, $63,000, 2022.
- NSERC Engage Grant, $24,995, 2017.
- C4 2018 Student Delegate Winner, $1,000, 2018.

SOFTWARE AND MODEL DEVELOPMENT

- Lead developer of `floodsr` v0.2.0, a Python CLI/PyTorch-GDAL package for AI super-resolution of flood hazard rasters using low-resolution grids and high-resolution DEMs.
- Lead developer of FloodDownscaler2 v2024-04-13, a Python CLI/library for flood-grid downscaling using CostGrow resampling, least-cost mapping, high-and-dry filtering, and WhiteboxTools.
- Lead developer of RICorDE v1.0.1, a PyQGIS/WhiteboxTools application for gridded water-depth estimation from flood inundation using HAND and cost-distance extrapolation.
- Lead developer of CanFlood v1.2.2 and CanFlood2 v2.1.0, Python QGIS plugins for Canadian flood-risk modeling, sensitivity analysis, multi-model workflows, and reporting.
- Lead developer of CanCurve v1.2.0, a Python QGIS plugin for creating Canadian flood depth-damage functions, with SQLite-backed project data and CanFlood-format exports.

PUBLICATIONS

Refereed Journal Articles and Preprints:

Buhrmann, Aaron, Nivedita Sairam, Cecilia I. Nievas, James E. Daniell, Heidi Kreibich, and Seth Bryant (Oct. 2026). "Building-Level Exposure Asset Value Modelling for Germany: An Ahrweiler Case Study". In: *Nat. Hazards Earth Syst. Sci.* 26.10, pp. 4785–4805. doi:10.5194/nhess-26-4785-2026.

Goulart, Henrique M. D., Lily-belle Sweet, Axel Deijns, Seth Bryant, and Benjamin Poschlod (2026). "Data-Driven Flood Classification to Assess Climate Change Effects on Riverine Compound Floods in Pre-Alpine Catchments". In: *EGUsphere [preprint]*. doi:10.5194/egusphere-2026-5735.

Bryant, Seth, Jody Reimer, Heidi Kreibich, and Bruno Merz (2025). "Aggregating Flood Damage Functions: The Peril of Jensen's Gap". In: *J. Flood Risk Manag.* 18.1, e13053. doi:10.1111/jfr3.13053.

Bryant, Seth, Jonas Wassmer, Paul Schimansky, Lindsay T. Keegan, Maria Pregnolato, Jürgen Kurths, Norbert Marwan, and Bruno Merz (Aug. 2025). "Unveiling Hidden Risks in Healthcare from Flood-Induced Transportation Disruption in Germany". In: *Commun. Earth Environ.* 6.1, pp. 1–11. doi:10.1038/s43247-025-02645-y.

Bryant, Seth, Guy Schumann, Heiko Apel, Heidi Kreibich, and Bruno Merz (Feb. 2024). "Technical Note: Resolution Enhancement of Flood Inundation Grids". In: *Hydrol. Earth Syst. Sci.* 28.3, pp. 575–588. doi:10.5194/hess-28-575-2024.

Bryant, Seth, Heidi Kreibich, and Bruno Merz (2023). "Bias in Flood Hazard Grid Aggregation". In: *Water Resour. Res.* 59.9, e2023WR035100. doi:10.1029/2023WR035100.

Updated October 28, 2026

Mester, Benedikt, Thomas Vogt, Seth Bryant, Christian Otto, Katja Frieler, and Jacob Schewe (Nov. 2023). "Human Displacements from Tropical Cyclone Idai Attributable to Climate Change". In: *Nat. Hazards Earth Syst. Sci.* 23.11, pp. 3467–3485. doi:10.5194/nhess-23-3467-2023.

Bryant, Seth, Evan G. R. Davies, David Sol, and Sandy Davis (Mar. 2022). "The Progression of Flood Risk in Southern Alberta since the 2013 Flood". In: *J. Flood Risk Manag.*, p. 18. doi:10.1111/jfr3.12811.

Bryant, Seth, Heather McGrath, and Mathieu Boudreault (Apr. 2022). "Gridded Flood Depth Estimates from Satellite-Derived Inundations". In: *Nat. Hazards Earth Syst. Sci.* 22.4, pp. 1437–1450. doi:10.5194/nhess-22-1437-2022.

PROCEEDINGS, CHAPTERS, REPORTS, AND SOFTWARE OUTPUTS

Bryant, Seth, Heidi Kreibich, and Bruno Merz (Apr. 2024). "Flood Damage Model Bias Caused by Aggregation". In: *Proceedings of IAHS*. Vol. 386. Copernicus GmbH, pp. 181–187. doi:10.5194/piahs-386-181-2024.

Bryant, Seth (2021). *CanFlood*. Natural Resources Canada.

Murdock, Heather and Seth Bryant (2021). *Federal Flood Risk Assessment Procedures v1.0*. Tech. rep. Natural Resources Canada and Public Safety Canada.

Bryant, Seth (Sept. 2018). "Modelling of Impact of Char Land Sedimentation". In: *Annex-F River Stabilization Impacts*. Ed. by Gerrit Klaassen and Erik Mosselman. Vol. 4. River Stabilization Plan.

Bryant, Seth and Evan G. R. Davies (Feb. 2018). "Living with Rivers". In: *Resilient Systems, Resilient Communities*. Edmonton, AB, pp. 58–83.

Bryant, Seth and Erik Mosselman (2017). "Taming the Jamuna: Effects of River Training in Bangladesh". In: *NCR Days 2017: Book of Abstracts*.

INVITED TALKS AND SEMINARS

Bryant, Seth (Sept. 2026a). *Floods at scale*. Computational Hydrology Group, Department of Civil Engineering, University of Calgary. Invited talk. Calgary, Alberta, Canada.

Bryant, Seth, Saurabh Kaushik, Jonathan Sullivan, and Beth Tellman (Dec. 2025). *Towards a multi-sensor satellite AI database of observed U.S. flood extents 2001–2024*. AGU Fall Meeting 2025. Invited talk.

Bryant, Seth (Nov. 6, 2024a). *Faking high resolution in flood maps*. Engineering Seminar, Utah State University. Invited talk.

Bryant, Seth (2023a). *Scaling Issues in Flood Risk Models*. RSS-Hydro Seminar. Invited talk.

Bryant, Seth (Apr. 3, 2023c). *Modelling in Flood Management*. Modelling Across Disciplines. Invited talk. Philadelphia, United States.

Bryant, Seth (Sept. 30, 2017). *Living with Rivers: Flood Management in Alberta*. RAIL Commons. Invited talk. 45-minute lecture. Olds, AB, Canada.

CONFERENCE PRESENTATIONS

Bryant, Seth (June 2026b). *Physics-guided super-resolution downscaling for national urban FIM*. 79th CWRA National Conference: Honouring the Spirit of Water. Winnipeg, Manitoba.

Bryant, Seth (June 3, 2026c). *Tracking 20 years of regulatory floodplain change from letters of map revision (LOMR)*. ASFPM 50th Annual National Conference, E9: Historical Perspectives for FEMA Mapping. Milwaukee, Wisconsin.

Bryant, Seth (May 2026d). *Implementation of physics-guided super-resolution downscaling for national urban FIM*. 4th Annual CIROH Developers Conference. Salt Lake City, Utah.

Bryant, Seth (June 2024b). *Resolution Enhancement for Flood Inundation*. Natural Hazards and Risks in a Changing World. Amsterdam, NL.

Bryant, Seth, Shahin Khosh Bin Ghomash, Heiko Apel, Bruno Merz, and Kasra Rafiezadeh Shahi (Dec. 14, 2023). *Deep learning-based super-resolution of flood inundation grids*. AGU Fall Meeting 2023. United States.

Bryant, Seth (July 2023b). *Resolution enhancement of flood inundation grids*. IUGG 2023. Berlin.

Bryant, Seth (Feb. 20, 2023e). *Flood damage model bias caused by aggregation*. 9th International Conference on Flood Management (ICFM9). Tsukuba, Japan.

Bryant, Seth (June 6, 2022). *CanFlood: a new open-source toolbox for flood risk modelling*. CWRA 2022 National Conference. Canmore, Canada.

Bryant, Seth, Heidi Kreibich, and Bruno Merz (May 24, 2022). *Flood damage model bias caused by aggregation*. EGU General Assembly 2022. Vienna, Austria. doi:10.5194/egusphere-egu22-5679.

Bryant, Seth (June 1, 2021). *Accumulating Flood Risk*. CWRA 2021 National Conference. Online.

Bryant, Seth and David Sol (July 24, 2020). *FLOOD RISK ASSESSMENT OF BC'S LOWER MAINLAND: Methods and Lessons*. Collaboratorium.

Bryant, Seth (Jan. 30, 2018). *ABMRI: An Approach to Dynamic Flood Risk Assessment*. C4 2018. Poster and Presentation. Gatineau, QC, Canada.

TEACHING AND SUPERVISION

2021 – 2024: Co-supervised an MSc student in Water Resources Engineering at the University of Alberta; thesis: *Quantifying Flood Depth: Remote Estimation of Flood Depths with Fast Response Tools* (doi:10.7939/r3-yxbr-7640).

2022 – 2024: Co-supervised an MSc student in Climate, Earth, Water, Sustainability (CLEWS) at the University of Potsdam; thesis: *Building Level Exposure Modelling for Flood Damage*.

2019 – 2021: Co-supervised an MEng student in Water Resources Engineering at the University of Alberta; thesis: *Impact of the 2020 Fort McMurray Flood on Local Businesses*.

2024: Teaching assistant, University of Potsdam, Grundpraktikum Geoökologie (Harz Field Course; Hydrological Measurement Methods). Ran the field program and prepared assignments and laboratory exercises.

2021 – 2022: Teaching assistant, University of Potsdam, Introduction to the Earth System (CLEWS CE01), Measurement and Observation Techniques. Organized lecture series activities, graded presentations, and coordinated with lecturers.

2017: Teaching assistant, University of Alberta, Environmental Hydrology ENVE 320. Graded assignments, tutored undergraduate students in hydrology and HEC-HMS, and delivered a lecture on flood risk management.

EXPERT ADVISORY COMMITTEES AND SERVICE

- Public Safety Canada's Technical Working Group on Floodplain Mapping and Flood Risk Assessment, 2019–current.
- Natural Resources Canada's Flood Hazard Mapping Standards Subgroup, 2021–current.
- Natural Resources Canada's Flood Hazard Identification and Priority Setting Guidelines Working Group, 2020–current.
- Public Safety Canada's Flood Risk Assessment Procedures Working Group, 2022–current.
- Canadian Coalition of Women in Engineering, Science, Trades and Technology Biennial Conference, volunteer scribe and moderator, 2018.
- Alpine Club of Canada, volunteer trip leader, 2015–2019.

REGISTRATIONS AND CERTIFICATIONS

- Professional Engineer, Association of Professional Engineers and Geoscientists of Alberta.
- Certified Professional in Erosion and Sediment Control, EnviroCert.
- LEED Green Associate, U.S. Green Building Council.

SELECTED PROFESSIONAL EXPERIENCE

2023 – current: Principal and President, CEFlood, Manitoba, Canada.

2015 – current: Hydrotechnical Engineer, REB Engineering, St. Helena, U.S.A. Civil and hydraulic design for floodplain, bridge, drainage, detention, and low-impact development projects.

2019 – 2022: Lead Flood Risk Modeller, IBI Group, Calgary, Canada. Led and supported flood risk modelling software development, risk assessments, and guideline development.

2020 – 2021: Flood Risk Analyst, Northwest Hydraulics Consulting, Vancouver, Canada.

2017 – 2019: River Engineer, Northwest Hydraulics Consulting, Dhaka, Bangladesh. Feasibility design and morphology analysis for Jamuna River training works.

2011 – 2014: Field Engineer and Civil Engineer, Worley Parsons, Edmonton, Alberta. Civil design, drainage, construction management, QA/QC, and field engineering for energy and infrastructure projects.

2007 – 2010: Civil Engineer and Summer Intern, REB Engineering, St. Helena, U.S.A. Civil site, grading, drainage, erosion control, survey, and winery infrastructure projects.

SELECTED PROFESSIONAL ACTIVITIES

2026: **Napa River hydraulic armoring design, Davis Property, St. Helena, California.** Prepared hydraulic background and rip-rap armor recommendations for a proposed riverbank repair after a winter 2024–2025 bank collapse, adapting a 1D HEC-RAS Napa River model, applying the FEMA 100-year discharge, estimating local bend velocities with USACE STREMR methods, and sizing revetment armor with the Maynord stability approach.

2025: **Residential flood AAL methodology and uncertainty review, Public Safety Canada.** Led a CE-Flood technical advisory scope to review Public Safety Canada's residential flood Average Annual Loss calculations, including the underlying assumptions, methods, code, and data. Work products include a project work plan, written feedback memo on AAL methodology and code, damage-curve use recommendation, sensitivity-analysis outline, and summary report.

2025: **Ehlers Lane culvert replacement hydraulic analysis, St. Helena, California.** Evaluated a failed private-driveway crossing over an ephemeral Napa River tributary using Napa County Road and Street Standards, USGS StreamStats peak-flow estimates, and FHWA HY-8 modeling, confirming that a proposed 6 ft reinforced-concrete culvert passes the 10-year and 100-year design floods without roadway overflow.

2024 – 2025: **CanCurve depth-damage function tool, Natural Resources Canada.** Proposed and led development of a QGIS plugin for creating user-specified flood depth-damage functions from NRCan datasets. The activity covered prototype review, software requirements, Qt interface wireframes, UML design, backend and frontend implementation, unit and integration testing, public QGIS plugin release planning, ReadTheDocs documentation, tutorials, user testing, issue tracking, and maintenance for an annual QGIS LTR update.

2024: **CanFlood modernization, Natural Resources Canada.** Scoped modernization of CanFlood v2.0 to preserve the core open-source modelling framework while improving the plugin's GUI, documentation, and user workflow. The modernization plan was guided by practitioner feedback and focused on a clearer, more maintainable QGIS user experience for typical flood-risk modelling workflows.

2023: **Calistoga Resiliency Center preliminary drainage analysis, Calistoga, California.** Developed pre- and post-development HEC-HMS rainfall-runoff models for a proposed industrial power-production facility in FEMA Zone AE, using NOAA Atlas 14 precipitation, NRCS soil conditions, detention storage, CULTEC chambers, and outlet works to show that peak post-development discharge could be limited to pre-development discharge.

2019 – 2022: **Canadian flood-risk software and guidance.** Served as lead developer and flood risk modeller for CanFlood phases 1–3, delivering an object-based, transparent, open-source QGIS plugin for Natural Resources Canada. The work expanded from core model building, execution, and visualization tools to levee-failure analysis, cost-benefit analysis, agricultural loss modelling, model sensitivity analysis, French documentation, and automated reporting. Also contributed to federal flood risk assessment procedures through literature review, advisory-committee material, meeting support, and guideline drafting.

2019 – 2022: **Regional and local flood-risk assessments in British Columbia.** Led or supported applied risk modelling for BC's Lower Mainland, the Okanagan Valley Watershed, and communities including Kent, Mission, New Westminster, and Langley. Activities included stakeholder-informed scope development, integration of rainfall-runoff, inundation, nearshore wave, and levee-breach hazards, asset inventory development, non-residential survey design, residential structure classification, dynamic multi-criteria decision support, and CanFlood-based consequence modelling.

2016 – 2019: **River training and hydrotechnical design, Bangladesh and North America.** Designed and evaluated Jamuna River training works for FRERMIP, including a 7 km hybrid revetment and sill, a recurring sand-plug channel-closure concept, dredge production estimates, Delft3D and SOBEK morphology modelling, and empirical sand-loss analysis. Earlier engineering activities included HEC-HMS and HEC-RAS hydrology and hydraulic studies, detention design, bridge hydraulic design, flood impact analysis, stormwater systems, culvert sizing, erosion control, grading, drainage, and construction QA/QC.